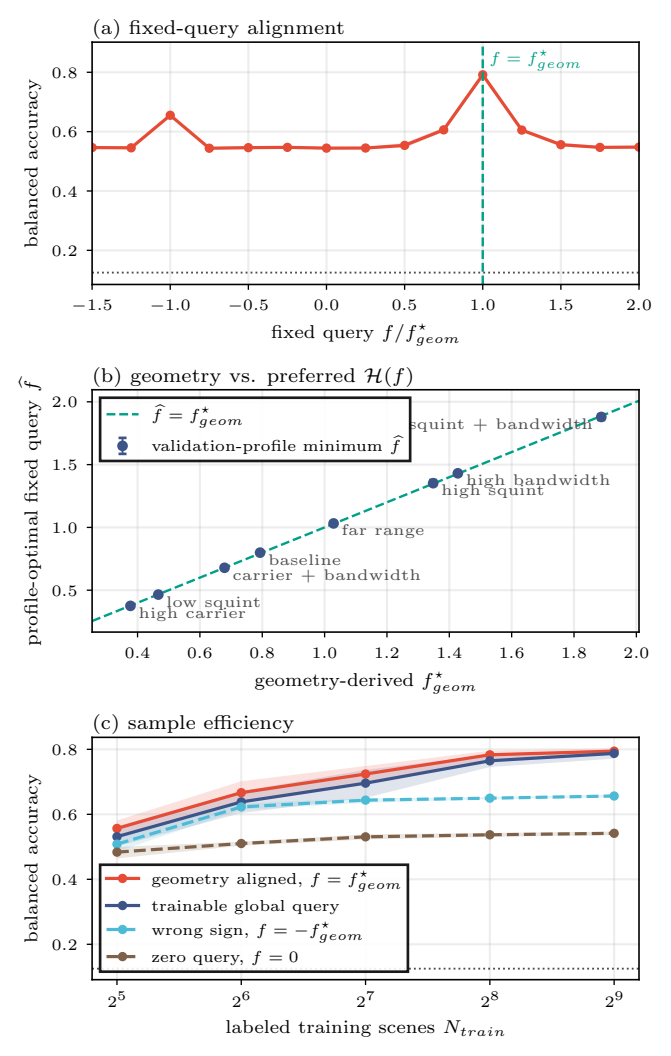

(a) fixed-query alignment
$f = f^\star_{geom}$
balanced accuracy
0.8
0.6
0.4
0.2
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
fixed query $f/f^\star_{geom}$
(b) geometry vs. preferred $\mathcal{H}(f)$
$\widehat{f} = f^\star_{geom}$
validation-profile minimum $\widehat{f}$
profile-optimal fixed query $\widehat{f}$
2.0
1.5
1.0
0.5
squint + bandwidth
high bandwidth
high squint
far range
baseline
carrier + bandwidth
low squint
high carrier
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
geometry-derived $f^\star_{geom}$
(c) sample efficiency
balanced accuracy
0.8
0.6
0.4
0.2
geometry aligned, $f = f^\star_{geom}$
trainable global query
wrong sign, $f = -f^\star_{geom}$
zero query, $f = 0$
2^5
2^6
2^7
2^8
2^9
labeled training scenes N_{train}